

SQL Hands-on Use Cases (21-25)

USE CASE 21: Create View

```
CREATE VIEW LatestCovidData AS
SELECT country, date, confirmed, deaths, recovered
FROM covid_cases c
WHERE date = (
    SELECT MAX(date)
    FROM covid_cases
    WHERE country = c.country
);

SELECT * FROM LatestCovidData;
```

USE CASE 22: Mortality Rate Function

```
DELIMITER $$

CREATE FUNCTION CalculateMortalityRate(
    p_country VARCHAR(100),
    p_date DATE
)
RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
    DECLARE mortality DECIMAL(10,2);

    SELECT (deaths/confirmed)*100
    INTO mortality
    FROM covid_cases
    WHERE country=p_country
    AND date=p_date;

    RETURN mortality;
END$$

DELIMITER ;

SELECT CalculateMortalityRate('India','2022-01-15');
```

USE CASE 23: Stored Procedure

```
DELIMITER $$

CREATE PROCEDURE UpdateCovidStats(
    IN p_country VARCHAR(100),
    IN p_date DATE,
    IN p_confirmed INT,
    IN p_deaths INT,
    IN p_recovered INT
)
BEGIN
    UPDATE covid_cases
    SET confirmed=p_confirmed,
        deaths=p_deaths,
        recovered=p_recovered
    WHERE country=p_country
    AND date=p_date;
END$$

DELIMITER ;

CALL UpdateCovidStats('India','2022-01-15',45000,520,43000);
```

USE CASE 24: Recovery Rate Function

```
DELIMITER $$

CREATE FUNCTION CalculateRecoveryRate(
    p_country VARCHAR(100)
)
```

```

RETURNS DECIMAL(10,2)
DETERMINISTIC
BEGIN
DECLARE recovery DECIMAL(10,2);

SELECT (recovered/confirmed)*100
INTO recovery
FROM covid_cases
WHERE country=p_country
ORDER BY date DESC
LIMIT 1;

RETURN recovery;
END$$

DELIMITER ;

SELECT CalculateRecoveryRate('India');
-----

```

USE CASE 25: Audit Trigger

```

CREATE TABLE covid_audit(
audit_id INT AUTO_INCREMENT PRIMARY KEY,
country VARCHAR(100),
update_date DATE,
old_confirmed INT,
new_confirmed INT,
old_deaths INT,
new_deaths INT,
old_recovered INT,
new_recovered INT,
changed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

DELIMITER \$\$

```

CREATE TRIGGER CovidAuditTrigger
BEFORE UPDATE
ON covid_cases
FOR EACH ROW
BEGIN
INSERT INTO covid_audit(
country, update_date,
old_confirmed,new_confirmed,
old_deaths,new_deaths,
old_recovered,new_recovered)
VALUES(
OLD.country,OLD.date,
OLD.confirmed,NEW.confirmed,
OLD.deaths,NEW.deaths,
OLD.recovered,NEW.recovered);
END$$

```

DELIMITER ;

```

UPDATE covid_cases
SET confirmed=60000,
    deaths=900,
    recovered=58000
WHERE country='India'
AND date='2022-01-15';

```

```

SELECT * FROM covid_audit;

```